MIL-I-631D
INTERIM AMENDMENT - II (SHIPS)
6 November 1962
SUPERSEDING
INTERIM AMENDMENT - I (SHIPS)
9 March 1962

MILITARY SPECIFICATION

INSULATION, ELECTRICAL, SYNTHETIC-RESIN COMPOSITION, NONRIGID

(This interim amendment forms a part of Military Specification MIL-I-631D dated 15 November 1961. It was developed by the Department of the Navy, Bureau of Ships, for immediate use pending the inclusion of its contents in a coordinated document.)

Page 2, paragraph 2.2: Delete reference to "D150" and "D876" and substitute:

"D150 - Methods of Test for A-C Capacitance, Dielectric Constant and Loss Characteristics of Electrical Insulating Materials.
"D876 - Methods of Testing Nonrigid Vinyl Chloride Polymer Tubing."

Page 3, paragraph 3.1.1.1, first sentence: Delete and substitute "Film is defined as material with a thickness of 0.010 inch or less, in widths greater than 3 inches."

Page 4, table I, column 9: For dielectric strength, delete "3,500" through "85" inclusive and substitute "3,500", "3,200", "3,000", "80", "80", and "85". For elongation, insert "50" between "50" and "35".

Pages 5 and 6, table II: Delete and substitute:

FSC 5970

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"Table II. —Property values of synthetic-resin electrical insulation—forms S and T (greater than 0. 002 inch in thickness).

| Property to be tested | Test reference | Conditioning procedure (see 4.4) | Unit of value | Nominal thickness | Value required for each type of material | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | A | B | C | D | E | F Grade a | G |
| Dielectric strength 1/ | 4.6.2 | C-96/23/0 | Volts per mil (minimum) | Inch | | | | | | | |
| | | | | 0.003 | 3,100 | 1,900 | 1,900 | 2,600 | 1,200 | ..... | 2,450 |
| | | | | .005 | 2,400 | 1,550 | 1,550 | 2,150 | 1,070 | ..... | 2,000 |
| | | | | .0075 | ..... | ..... | ..... | ..... | ..... | ..... | 1,600 |
| | | | | .008 | 1,800 | 1,300 | 1,250 | 1,800 | 960 | 1,045 | ..... |
| | | | | .010 | 1,600 | 1,200 | 1,150 | 1,550 | 920 | ..... | 1,500 |
| | | | | .012 | ..... | ..... | ..... | ..... | ..... | 950 | ..... |
| | | | | .014 | ..... | ..... | ..... | ..... | ..... | ..... | 1,400 |
| | | | | .016 | 1,300 | 1,000 | 950 | 1,200 | 940 | 820 | ..... |
| | | | | .020 | 1,100 | 900 | 840 | 1,050 | 790 | 700 | ..... |
| | | | | .025 | 960 | 820 | 760 | 920 | 740 | 630 | ..... |
| | | | | .030 | 870 | 760 | 700 | 830 | 700 | 570 | ..... |
| | | | | .040 | 740 | 640 | 620 | 700 | 600 | 480 | ..... |
| | | | | .050 | 660 | 550 | 550 | 610 | 530 | 430 | ..... |
| | | | | .060 | 600 | 490 | 490 | 540 | 450 | 400 | ..... |
| Dielectric strength 1/ | | C-96/23/96 | Percent of dry value obtained on test (min.) | All thicknesses | 75 | 85 | 75 | 55 | 90 | 90 | 90 |
| Dielectric strength 1/ | | C-96/23/96 | Percent of dry value obtained on test (min.) | Inch | | | | | | | |
| | | | | 0.003 | ..... | ..... | ..... | ..... | ..... | ..... | 85 |
| | | | | .005 | ..... | ..... | ..... | ..... | ..... | ..... | 90 |
| | | | | .0075 | ..... | ..... | ..... | ..... | ..... | ..... | 90 |
| | | | | .010 | ..... | ..... | ..... | ..... | ..... | ..... | 90 |
| | | | | .014 | ..... | ..... | ..... | ..... | ..... | ..... | 90 |
| Dissipation factor (form S only) | | | | | Natural colors / Other colors | | | | | | |
| At 1 kilocycle | 4.6.3 | C-96/23/0, C-96/23/96 | Maximum | ..... | 0.0006 / 0.0025 | 0.02 | 0.035 | 0.015 | 0.08 | 0.14 | 0.006 |
| | | | | | .0008 / .0040 | .03 | .070 | .03 | .08 | .16 | .0065 |
| At 1 megacycle | | C-96/23/0, C-96/23/96 | Maximum | ..... | .0006 / .0025 | .045 | .05 | .028 | .... | .... | .020 |
| | | | | | .0008 / .0040 | .055 | .07 | .045 | .... | .... | .025 |

1/ For thickness not shown, the dielectric strength shall be that given for the nearest thickness shown."



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"Table II. —Property values of synthetic-resin electrical insulation—forms S and T (greater than 0.002 inch in thickness) (cont'd)

| Property to be tested | Test reference | Conditioning procedure (see 4.4) | Unit of value | Nominal thickness | Value required for each type of material | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | A | | B | C | D | E | F Grade a | G |
| | | | | | Natural colors | Other colors | | | | | | |
| Dielectric constant (form S only) | | | | | | | | | | | | |
| At 1 kilocycle | 4.6.3 | C-96/23/0 | Maximum | ..... | 2.5 | 2.5 | 4.0 | 4.5 | 4.0 | 5.0 | 7.0 | 3.5 |
| | | C-96/23/96 | | | 2.7 | 2.7 | 5.0 | 8.7 | 5.0 | 6.0 | 8.0 | 3.7 |
| At 1 megacycle | | C-96/23/0 | Maximum | ..... | 2.5 | 2.5 | 4.0 | 4.3 | 4.0 | ..... | | 3.3 |
| | | C-96/23/96 | | | 2.7 | 2.7 | 5.0 | 7.0 | 5.0 | ..... | | 3.5 |
| Volume resistivity (form S only) | 4.6.4.1 | C-96/23/0 | Ohm-centimeters (min.) | ..... | $10^{17}$ | | $10^{13}$ | $10^{13}$ | $10^{14}$ | $10^{13}$ | $10^{11}$ | $10^{15}$ |
| | | C-96/23/96 | | | $10^{16}$ | | $10^{12}$ | $10^{10}$ | $12^{12}$ | $10^{12}$ | $10^{10}$ | $10^{15}$ |
| Tensile strength | 4.6.5.1 | C-96/23/50 | P.s.i. (min.) | ..... | 1,400 | | 5,000 | 6,000 | 3,500 | 3,500 | 1,800 | 17,000 |
| | | | | .014 | ..... | | ..... | ..... | ..... | ..... | ..... | 16,000 |
| Elongation | 4.6.5.1 | C-96/23/50 | Percent (min.) | .010 | 350 | | 25 | 10 | 25 | 15 | 200 | 80 |
| | | | | .014 | ..... | | ..... | ..... | ..... | ..... | ..... | 100 |
| | | C-96/23/50 | Percent (max.) | ..... | ..... | | 100 | 50 | 100 | 100 | 450 | ..... |
| | | C-96/23/50 | | .014 | ..... | | ..... | ..... | ..... | ..... | ..... | |
| | | E-400/100 | Percent change (max.) | ..... | ..... | | ..... | ..... | ..... | ..... | 35 | ..... |
| Elongation | | E-400/130 | Percent change (max.) | ..... | ..... | | ..... | ..... | ..... | ..... | ..... | 25 |
| | | | | .010 | ..... | | ..... | ..... | ..... | ..... | ..... | 25 |
| | | | | .014 | ..... | | ..... | ..... | ..... | ..... | ..... | 30 |
| Percent return | 4.6.5.2 | C-96/23/50 | Percent (min.) | ..... | ..... | | ..... | ..... | ..... | ..... | 80 | ..... |
| Softening temperature (penetration) | 4.6.7 | C-96/23/50 | Degrees centigrade (min.) | ..... | 65 | | 120 | 160 | 110 | 120 | 75 | 200 |
| Lengthwise shrinkage | 4.6.8 | E-2/100 | Percent (max.) | ..... | 2.0 | | 3.0 | 3.0 | 2.0 | 5.0 | 10.0 | ..... |
| | | E-2/150 | Percent (max.) | ..... | ..... | | ..... | ..... | ..... | ..... | ..... | 5.0 |

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Page 10, table IV, column 2: Opposite ". 0015" insert ". 0004".

Page 10, table V: Opposite ". 010" insert ". 0016" under type G. Between ". 012" and ". 016" insert ". 014" under nominal thickness and ". 0020" under type G.

Page 11, table VII: Delete and substitute:

"Table VII - Tubing lengths.

| Nominal inside diameter | Minimum length |
|---|---|
| Inch | Feet |
| 0.022 - 0.027 | 1,200 |
| .034 - .059 | 1,000 |
| .066 - .166 | 500 |
| .186 - .330 | 200 |
| .360 - .600 | 100 |
| .600 and over | 50 |

Page 14, paragraph 4.3.1: Delete and substitute:

"4.3.1 Samples. - Separate samples for each type, form, and subform of material for which approval is requested shall be furnished in sizes and quantities shown in table X (see 4.3.2.2). Qualification for forms $U_a$ and $U_b$ may be obtained separately because of the large number of sizes in these subforms. The same applies to both subforms ($T_f$ and $T_s$) under form T. If all of the thicknesses or sizes shown in table X are not tested, the qualification will apply only to the thickness or size tested."

Preparing activity:
Navy - Ships
(Project 5970-N043Sh)



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